Association and Causality

APHS 20203 - Epidemiology & Biostatistics

Epidemiological Study: Fundamental Question

After identifying patterns (Person–Place–Time), we ask:

- **Are exposure and outcome linked?**
- Is the observed association valid or misleading?
- Could chance, bias, or confounding explain the finding?
- Does the evidence support a causal interpretation?

Disease Causality: Concepts Over Time

Causal explanations evolve with **available evidence, tools, and methods**.

- **Supernatural explanations:** Disease attributed to fate, spirits, or moral forces
- **Environmental (Miasma) theories:** Illness linked to air, water, climate, and surroundings
- **Germ theory:** Specific biological agents identified as causes of many infectious diseases
- **Risk-factor & multicausal models:** Most diseases arise from **multiple interacting factors**, not a single cause
- **Modern epidemiology:** Causal inference integrates biology, behavior, environment, and social context over time

What Do Epidemiologists Report First?

When studying health in populations, epidemiologists usually start by reporting:

A. Clear cause-and-effect relationships

B. Patterns and correlations

C. Biological mechanisms

D. Random events with no structure

Association

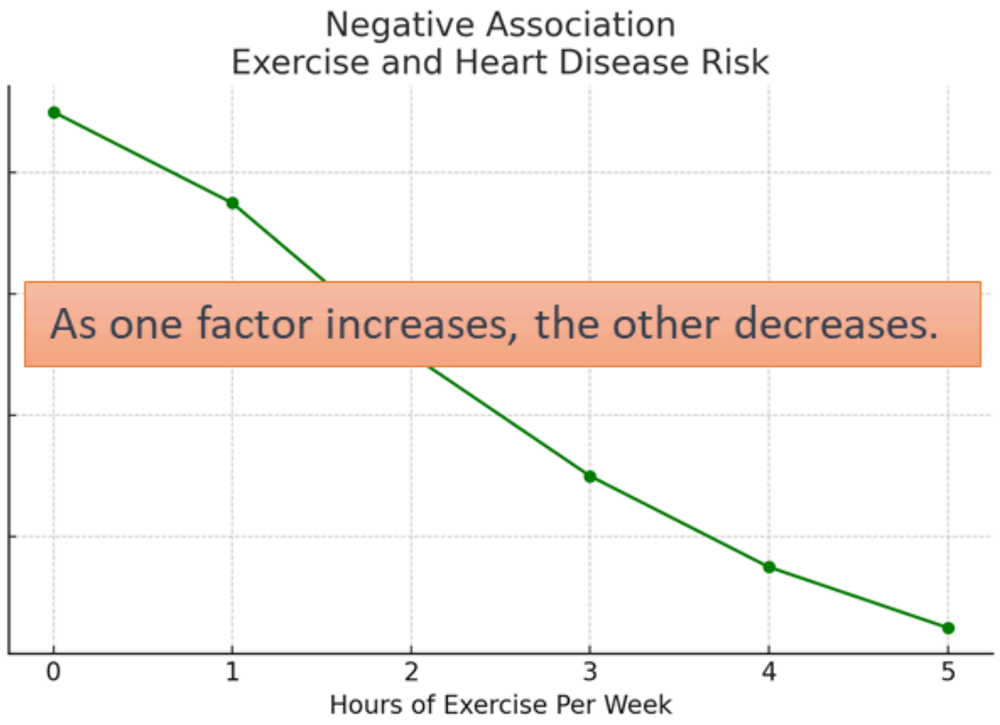
- **Association:** The distribution of an outcome differs across levels of an exposure.
  - For example, the outcome may be more common in one group than another.
- Knowing something about **X** helps us predict something about **Y**.
- **Types:** Positive association and negative association.
- An association alone does not tell us **why** the groups differ.

Types of Association

Positive Association



Negative Association



Causality

- **Causality:** A relationship where a change in one factor **directly produces** a change in an outcome.
- **Key distinctions**
  - **Causality implies association**
  - **Association does not imply causality**
- **Why causality is harder to establish**
  - Multiple contributing factors
  - Role of chance, bias, and confounding
  - Ethical and practical limits on experimentation

The Search for Causality

- When an association is observed, epidemiologists may ask:
  - **Is the association causal?**
    Could chance, bias, or confounding explain it?
  - **Is the association consistent with what we (think we) know about how the world works?**
    Across time, place, populations, and biology?
  - **Does the evidence support a causal interpretation?**
    Or are alternative explanations more plausible?
- **Key idea:** Causality is not proven by a single study, but **evaluated through accumulated evidence**.

Bias

- **Bias:** *Systematic* error in an estimate. The gap between what we measured and what we intended to measure.
- Bias can arise from **who enters a study**, **how we measure variables**, or **how we estimate a causal effect**.
- **Example:** We want to estimate average physical activity among all Texas adults, so we measure physical activity patterns at local gyms.
- **Key idea:** A larger sample does not generally eliminate bias.

Confounding

- **Confounding:** The effects of other factors become mixed into our estimate of an exposure's causal effect. Or giving credit to the wrong variable.
- **Example:** Ice cream sales and drownings both increase during hot weather.
  - Hot weather influences both ice cream sales and drowning risk.
- **Question:** Would reducing ice cream sales reduce drownings?
- **Key idea:** The association can be real and useful for prediction without representing a causal effect of ice cream sales.

Exercise: Sunscreen Use and Skin Cancer

- **Observed association:** Higher rates of skin cancer are associated with increased sunscreen use.
- **Question:** What might **confound** the relationship between sunscreen use and skin cancer?
- **Think about:**
  - Who is more likely to use sunscreen?
  - What other exposure might be related to both sunscreen use and skin cancer?

Exercise: Which Is a Spurious Association?

Which scenario shows a **spurious association**?

A. As outside temperature drops, more people wear hats.

B. As vaccination rates increase, the targeted disease decreases.

C. Fires attended by more firefighters tend to cause more property damage.

D. People who exercise regularly have lower rates of heart disease.

Population, Sample & Uncertainty

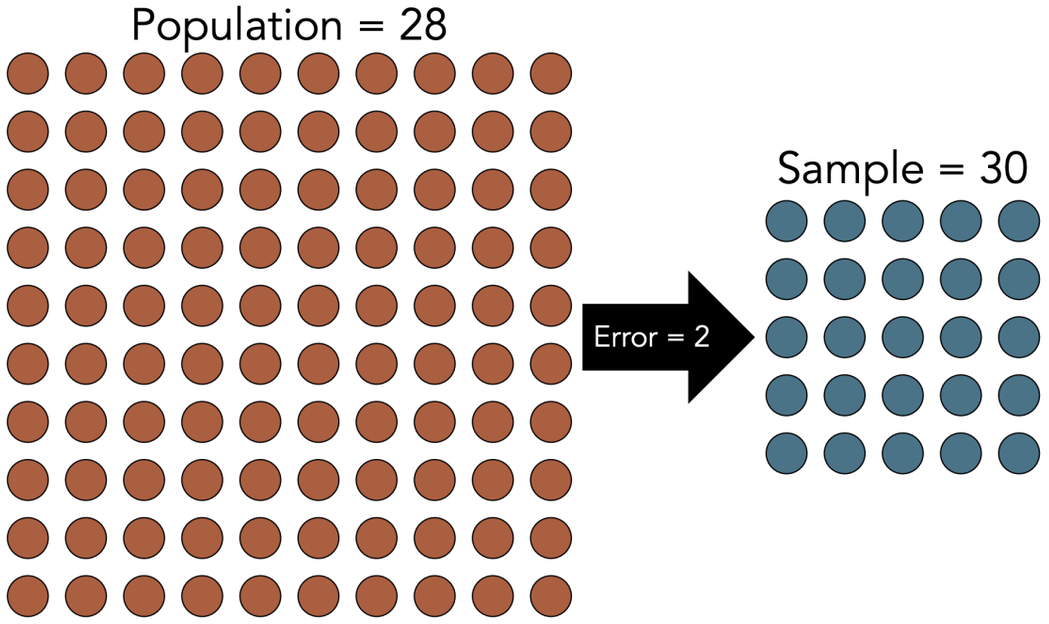
Example: We want to understand the health of adults living in Texas.

Group	Who belongs to it?	In our example
Target population	People we want findings to apply to	All Texas adults
Source population	People from whom we select participants	Adult patients at several Texas clinics
Study sample	People whose data we analyze	500 participating patients

- **Chance:** Would another random sample from the same source give a different estimate?
- **Applicability:** How well do these findings apply to all Texas adults?

Role of Chance in Association/Causality

Sampling variability: Estimates differ from sample to sample, even when we sample randomly from the same population.



Hypothetical BMI example

- Population mean: **28**
- One random sample: **30**
- Another random sample might have a mean of **26**.

Question: Must a difference between an estimate and the truth mean someone made a mistake?

Controlling for Chance in Epidemiologic Studies

- **Larger samples** generally reduce sampling variability.
- **Systematic bias is a different problem.** A larger sample does not typically correct biased recruitment, measurement, or confounding.
- **Precision and causality are different questions.** We can estimate an association precisely and still be uncertain whether it is causal.
- **Key idea:** More data can improve precision, but causal interpretation still requires careful judgment.

Types of Causality

- **Deterministic causality**
  - A cause leads to an outcome **with certainty**
  - If the cause occurs, the outcome follows
  - *Everyday example:* Pressing a light switch → the light turns on
- **Probabilistic causality**
  - A cause **changes the chance** of an outcome
  - The outcome does not always occur
  - *Everyday example:* Studying more → higher chance of doing well on an exam
  - Epidemiology focuses mostly on **probabilistic causality**.

Deterministic Causality in Epidemiology (Rare)

- **Defining feature:** Exposure leads to disease **with near certainty**
- **Why it's unusual**
  - Most diseases involve **multiple interacting causes**
  - Individual susceptibility and context matter
- **Epidemiologic example**
  - Untreated rabies virus exposure → rabies disease
- **Key takeaway:** Deterministic causality exists in epidemiology, but it is **rare and exceptional**.

Probabilistic Causality in Epidemiology (Typical)

- **Defining feature:** Exposure **increases the likelihood** of disease
- **Why this is typical**
  - Diseases arise from **multiple interacting causes**
  - Susceptibility varies across individuals
- **Epidemiologic example**
  - Smoking → higher risk of lung cancer
- **Key takeaway:** Most epidemiologic causal relationships involve **risk, not certainty**.

Exercise: Which Best Describes Deterministic Causality?

Which example best fits **deterministic causality**?

A. A medication leads to different levels of improvement across patients

B. A specific bacterium always causes disease in people without immunity

C. Increased exercise in a population is linked to lower heart disease rates

D. Rising global temperature is correlated with rising sea levels

Multivariate Causality

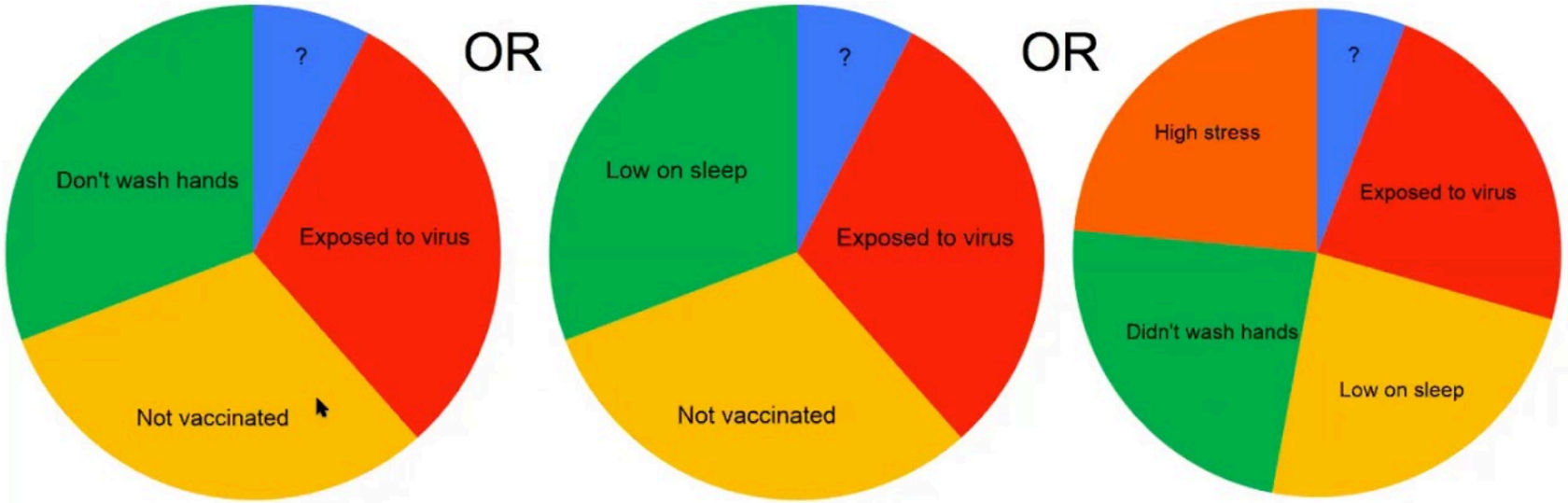
- Most health outcomes involve **multiple contributing causes**.
- Causes can act together, and different combinations can produce the same outcome.

Return to the light switch: What else must be present for flipping the switch to turn on the light?

- Electricity, a working bulb, and a complete circuit.
- Flipping the switch may complete the combination of conditions needed for the light to turn on.

Key idea: The final event in a sequence is not necessarily the only cause.

Rothman's Model of Sufficient and Component Causes (RMSCC)



Outcome: Flu

- Each pie is a **sufficient cause set** - "sufficient" to produce the outcome
- Each slice represents a **component cause**
- Different combinations of slices can produce the **same outcome**
- A **necessary cause** is a component cause that appears in every sufficient cause set

Project Prep 2

Quick Review: Association vs Causality

What distinguishes a **causal relationship** from an **association**?

A. A causal relationship means the two factors increase at the same time.

B. A causal relationship means changes in one factor lead to changes in another, with evidence that the exposure occurs before the outcome.

C. An association requires a statistically significant p-value, while a causal relationship does not.

D. Associations can only be observed in laboratory settings, while causal relationships are observed in real-world settings.

Hill's Viewpoints for Causality

- **Context:** Proposed by Sir Austin Bradford Hill (1965) to guide causal reasoning in epidemiology
- **Purpose:** A set of nine principles to consider when evaluating whether an observed **association** is likely to be **causal**
- **Important clarifications**
  - Not a checklist
  - Not all criteria must be met
  - **Temporality** (exposure precedes outcome) is essential
- **Key idea:** Hill's guidelines **organize evidence**; they do not prove causation.

1. Strength of Association (Effect Size)

A small association does not mean that there is not a causal effect, though the larger the association, the more likely that it is causal.

- **Example:** People who smoke cigarettes are about 25 times as likely to develop lung cancer as people who do not smoke (U.S. Department of Health and Human Services 2014).
- **Limitations:** Strong associations can still arise from confounding or bias. Effect size alone does not establish causation.
- **Counterexample:** Children's shoe sizes and reading aptitude are associated. What could explain both?

2. Consistency (Reproducibility)

Consistent findings across studies, populations, and settings strengthen the likelihood of a causal effect.

- **Example:** Higher sodium intake is associated with higher blood pressure across populations in many countries (Elliott et al. 1996).
- **Limitation:** Studies that share the same bias can produce consistently misleading findings.
- **Counterexample:** Suppose studies in many cities consistently find that ice cream sales and murders rise together. Does repeating this finding mean that ice cream causes murder?

3. Specificity

An association concentrated in a particular exposure–disease relationship can support a causal interpretation, but specificity is not required.

- **Example:** A disease-causing expansion of CAG (cytosine–adenine–guanine) repeats in the *HTT* gene causes Huntington's disease—a specific genetic alteration linked to a particular disease (National Library of Medicine 2020).
- **Limitations:** One exposure can cause several diseases, and one disease can have several causes. A lack of specificity does not rule out causation.
- **Counterexample:** Does the fact that smoking causes lung cancer make it any less likely that it also causes heart disease?

4. Temporality

The effect has to occur after the cause (and if there is an expected delay between the cause and expected effect, then the effect must occur after that delay).

- **Example:** Smoking precedes lung cancer development.
- **Limitations:** The order of exposure and outcome may be difficult to establish. Even when an exposure clearly precedes an outcome, that sequence alone does not mean it caused the outcome.
- **Counterexample:** A rooster crows before sunrise. Does it cause sunrise?
- **Key idea:** Temporality is necessary, but timing alone does not establish causality.

5. Biological Gradient (Dose-Response)

Increasing exposure is associated with increasing risk of the outcome, which can support a causal interpretation.

- **Example:** Smoking more cigarettes per day is associated with higher lung cancer risk.
- **Limitations:** Threshold effects and nonlinear relationships.
- **Counterexample:** Suppose days with higher ice cream sales also have more drownings. Does buying more ice cream increase the risk of drowning?

6. Biological Plausibility

A plausible mechanism between cause and effect increases the likelihood of a causal effect.

- **Example:** Chemicals in tobacco smoke can damage DNA in lung cells. That damage can disrupt the control of cell growth and lead to cancer (Centers for Disease Control and Prevention 2024).
- **Limitation:** What seems biologically plausible depends on current knowledge. A convincing explanation can be wrong, and a real causal effect may exist before its mechanism is understood.
- **Counterexample:** Miasma theory of disease causation.

7. Coherence

Coherence between epidemiological and laboratory findings increases the likelihood of a causal effect.

- **Example:** Smoking-related lung cancer aligns with known carcinogenic effects of tobacco.
- **Limitations:** Agreement with existing knowledge does not rule out alternative explanations. Apparent disagreement may reflect gaps or errors in current knowledge rather than the absence of a causal effect.
- **Counterexample:**
  - Epidemiology: Observational studies found lower coronary heart disease risk among hormone users (Grodstein et al. 1996).
  - Biology: Hormone therapy lowered LDL cholesterol, supporting a possible protective mechanism (The Writing Group for the PEPI Trial 1995).
  - Later evidence: The Women's Health Initiative randomized trial found that the estrogen-plus-progestin regimen tested did **not** prevent coronary heart disease and could increase its risk (Manson et al. 2003).
- **Comparison:** Consistency compares studies. Coherence compares findings with existing knowledge.

8. Experiment

Well-designed randomized experiments can strengthen causal inference by changing exposure and reducing confounding.

- **Example:** In the DASH-Sodium trial, participants consumed diets with different sodium levels in random order. Lower sodium intake reduced blood pressure (Sacks et al. 2001).
- **Limitations:** Experiments may be unethical or impractical. Missing outcomes, poor adherence, and measurement problems can undermine results. Findings may not generalize beyond the participants and conditions studied.
- **Counterexample:** Suppose a randomized treatment trial loses follow-up data from participants whose health worsens. Could the treatment appear more effective than it really is?

9. Analogy

A known causal relationship can suggest that a similar exposure might produce a similar outcome.

- **Example:** Rubella infection during pregnancy can cause birth defects. By analogy, another viral infection during pregnancy—such as Zika—might also harm fetal development (Rasmussen et al. 2016).
- **Limitations:** Similarities may be superficial. Similar exposures can act through different mechanisms and produce different outcomes. Analogy suggests a hypothesis; it does not establish causation.
- **Counterexample:** Smoking and getting punched in the gut can both make someone cough. Does that mean both cause lung cancer?

Best Practices for Thinking About Hill's Viewpoints

- **Start with temporality:** If exposure does not precede outcome, causality cannot be established.
- **Evaluate criteria collectively:** No single criterion is sufficient. Temporality is necessary, but it is not sufficient.
- **Consider alternative explanations:** Always assess confounding, bias, and chance.
- **Avoid checklist thinking:** Hill's viewpoints guide reasoning but do not prove causation.

The Search for Causality: EVALI Outbreak Example

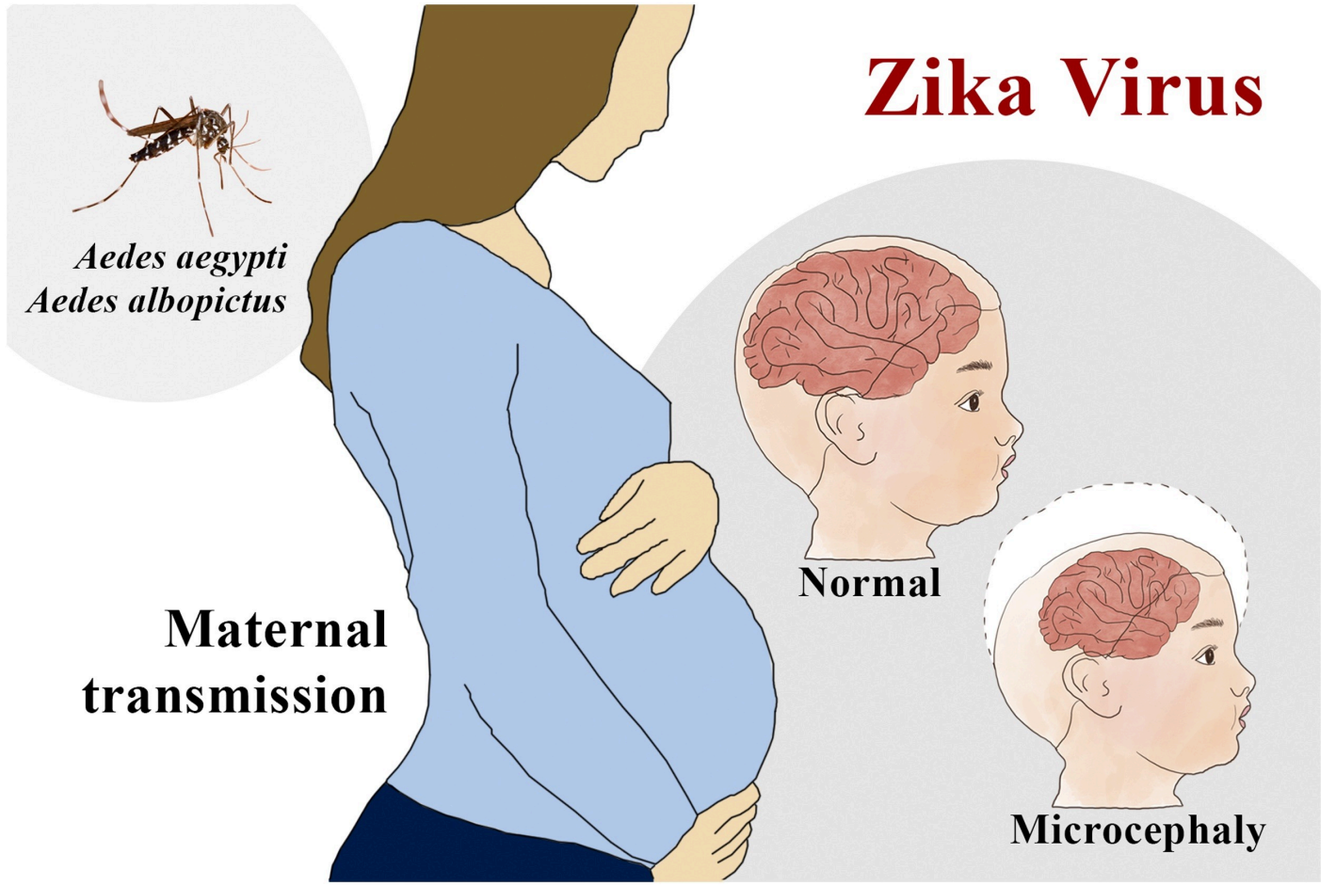
EVALI: E-cigarette or Vaping–Associated Lung Injury

Question: How did epidemiologists move from **observation** to **causal judgment**?

- **Observation (Early 2019):** Surge in lung injuries among people who reported vaping
- **Initial hypotheses:** Vaping products may be linked to lung injury: Which products? Which components?
- **Data collection (Mid-2019):** Patient histories, product testing, pattern comparison
- **Evidence accumulation (Late 2019):** Vitamin E acetate identified in many THC-containing products
- **Causal inference (2020):** Evidence supports a causal role for vitamin E acetate in EVALI

Example: Zika Virus & Microcephaly

- **Zika virus** is a member of the virus family Flaviviridae. It is spread by daytime-active Aedes mosquitoes.
- **Microcephaly** is a medical condition involving a smaller-than-normal head.



Group Activity: Zika Virus and Causal Reasoning

- You will watch a short video about the Zika virus and microcephaly.
- Each group has been assigned a **different lens** for evaluating causality at this Google link: https://tinyurl.com/W5GroupActivity or via the QR code
- **While watching the video:**
  - Focus only on your assigned lens and the last slide about dose-response
  - Capture brief bullet points, not full sentences
- **After the video, your group should report:**
  - One piece of evidence that supports causality
  - One limitation or uncertainty related to your lens
- **As a whole class, we will then discuss:**
  - Whether there is evidence of a dose–response relationship, and
  - Why some causal criteria may be harder to demonstrate than others



Video: Zika Virus & Microcephaly

Video player configuration error

Error 153

Watch video on YouTube Learn more

Error 153

Causality Criteria: Zika Virus & Microcephaly

- **Temporality:** Pregnant mothers got Zika before having babies with microcephaly.
- **Strength:** Big spike in microcephaly where Zika outbreaks occurred.
- **Consistency:** Similar rises reported in multiple regions (e.g., Brazil, French Polynesia).
- **Specificity:** Most increases appeared where Zika was active (though microcephaly has other causes).
- **Dose-Response (Biological Gradient):** More severe or earlier Zika infections may lead to higher risk of microcephaly.
- **Biological Plausibility:** Zika infects developing brain cells, a direct biological link to smaller head size.
- **Coherence:** Fits with existing knowledge that viruses can damage fetal brain development.
- **Experiment:** No direct clinical trials (ethical limits), but strong observational and lab evidence.
- **Analogy:** Similar viruses (e.g., rubella) cause birth defects, supporting the Zika-microcephaly link.

Putting It All Together

- Causal inference in epidemiology requires careful judgment.
  - Associations are typically reported first.
  - Causality is evaluated using multiple lines of evidence.
  - Most diseases have **multiple contributing causes**.
  - Uncertainty and chance are always present.
- **Final takeaway:** Epidemiology is about making the **best possible causal judgments** with imperfect evidence.

References

Centers for Disease Control and Prevention. 2024. “Health Effects of Cigarettes: Cancer.” September 17. https://www.cdc.gov/tobacco/about/cigarettes-and-cancer.html.

Elliott, Paul, Jeremiah Stamler, Rob Nichols, et al. 1996. “Intersalt Revisited: Further Analyses of 24 Hour Sodium Excretion and Blood Pressure Within and Across Populations.” *BMJ* 312 (7041): 1249–53. https://doi.org/10.1136/bmj.312.7041.1249.

Grodstein, Francine, Meir J. Stampfer, JoAnn E. Manson, et al. 1996. “Postmenopausal Estrogen and Progestin Use and the Risk of Cardiovascular Disease.” *New England Journal of Medicine* 335 (7): 453–61. https://doi.org/10.1056/NEJM199608153350701.

Kuadkitkan, Atichat, Nitwara Wikan, Wannapa Sornjai, and Duncan R. Smith. 2020. “Zika Virus and Microcephaly in Southeast Asia: A Cause for Concern?” *Journal of Infection and Public Health* 13 (1): 11–15. https://doi.org/10.1016/j.jiph.2019.09.012.

Manson, JoAnn E., Judith Hsia, Karen C. Johnson, et al. 2003. “Estrogen Plus Progestin and the Risk of Coronary Heart Disease.” *New England Journal of Medicine* 349 (6): 523–34. https://doi.org/10.1056/NEJMoa030808.

National Library of Medicine. 2020. “Huntington’s Disease.” MedlinePlus Genetics, July 1. https://medlineplus.gov/genetics/condition/huntingtons-disease/.

Rasmussen, Sonja A., Denise J. Jamieson, Margaret A. Honein, and Lyle R. Petersen. 2016. “Zika Virus and Birth Defects—Reviewing the Evidence for Causality.” *New England Journal of Medicine* 374: 1981–87. https://doi.org/10.1056/NEJMsr1604338.

Sacks, F. M., L. P. Svetkey, W. M. Vollmer, et al. 2001. “Effects on Blood Pressure of Reduced Dietary Sodium and the Dietary Approaches to Stop Hypertension (DASH) Diet.” *New England Journal of Medicine* 344 (1): 3–10.

https://doi.org/10.1056/NEJM200101043440101.

The Writing Group for the PEPI Trial. 1995. “Effects of Estrogen or Estrogen/Progestin Regimens on Heart Disease Risk Factors in Postmenopausal Women: The Postmenopausal Estrogen/Progestin Interventions (PEPI) Trial.” *JAMA* 273 (3): 199–208. https://pubmed.ncbi.nlm.nih.gov/7807658/.

U.S. Department of Health and Human Services. 2014. *The Health Consequences of Smoking—50 Years of Progress: A Report of the Surgeon General*. U.S. Dept of Health and Human Services. https://www.ncbi.nlm.nih.gov/books/NBK179276/.